

TORP, Norway

WMO: 014830

Lat: 59.184N

Lon: 10.255E

Elev: 88

StdP: 100.27

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.0	(c) -11.9	(d) -16.8	(e) 0.9	(f) -12.6	(g) -14.8	(h) 1.0	(i) -10.2	(j) 12.3	(k) 3.6	(l) 11.3	(m) 3.9	(n) 3.0	(o) 0	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	25.9	16.9	23.8	16.5	21.9	15.7	18.7	22.9	17.7	21.5	17.0	20.2	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	19.0	16.8	12.1	18.6	15.9	11.4	17.9	53.3	23.4	50.4	21.6	47.9	19.5	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.5	9.2	8.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-15.5	27.6	2.6	2.8	-17.3	29.6	-18.9	31.2	-20.3	32.7	-22.2	34.7
			-15.7	20.0	2.6	1.1	-17.6	20.8	-19.1	21.5	-20.6	22.1	-22.4	22.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-2.3	-1.8	1.9	6.3	11.1	14.6	17.1	15.5	12.4	7.1	2.7	-0.7		
	DBStd	7.68	4.70	4.63	4.06	3.34	3.77	2.69	2.57	1.86	2.48	3.21	3.90	5.29		
	HDD10.0	1773	382	329	253	121	30	1	0	8	100	218	332			
	HDD18.3	4146	640	562	510	362	224	118	57	89	177	347	468	590		
	CDD10.0	697	0	0	2	8	66	138	219	171	81	11	1	0		
	CDD18.3	27	0	0	0	0	2	6	18	2	0	0	0	0		
	CDH23.3	208	0	0	0	0	17	48	137	6	0	0	0	0		
	CDH26.7	31	0	0	0	0	1	4	26	0	0	0	0	0		
(14) Wind	WSAvg	3.7	3.8	3.8	3.8	3.6	3.7	3.6	3.4	3.4	3.6	3.8	4.0	3.9		
(15) Precipitation	PrecAvg	974	85	60	48	53	63	68	79	118	92	112	111	85		
	PrecMax	1364	189	185	100	115	153	131	175	227	283	312	212	182		
	PrecMin	726	14	7	1	14	2	5	12	33	38	7	18	27		
	PrecStd	178	46	42	30	27	38	31	35	49	62	70	49	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	10.0	16.2	19.0	25.1	27.0	29.2	24.2	20.2	15.8	12.0	10.0		
		MCWB	5.7	6.0	9.6	11.4	15.5	16.0	18.3	17.2	15.0	12.3	10.5	9.5		
	2%	DB	6.2	6.9	12.0	16.6	22.9	24.9	27.0	22.2	19.0	14.1	10.2	8.2		
		MCWB	5.1	3.6	6.7	9.7	14.7	15.9	18.2	16.4	14.4	11.9	9.4	7.6		
	5%	DB	5.1	5.1	9.1	14.2	20.1	22.0	24.8	20.9	17.8	13.1	9.1	7.1		
		MCWB	4.0	3.6	5.0	8.8	13.4	14.6	17.4	16.0	13.9	11.3	8.4	6.2		
	10%	DB	4.0	4.0	7.1	12.2	18.0	20.0	22.8	19.8	16.8	12.1	8.1	6.0		
		MCWB	3.0	3.1	3.7	7.6	12.5	13.5	16.8	15.3	13.7	10.4	7.4	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	6.5	10.0	12.0	17.6	18.2	20.0	19.2	18.0	14.0	11.2	9.4		
		MCDB	6.8	8.6	16.5	15.8	23.1	22.6	24.6	21.1	18.5	14.7	11.4	9.4		
	2%	WB	5.4	4.8	7.3	10.5	16.1	17.0	19.1	17.9	16.2	13.0	9.7	7.9		
		MCDB	5.9	5.5	10.7	15.0	20.2	22.3	24.3	20.4	17.2	13.6	10.0	8.3		
	5%	WB	4.3	4.0	5.7	9.5	14.8	16.0	18.4	17.1	15.2	12.0	8.6	6.3		
		MCDB	4.8	4.7	7.5	13.6	18.2	19.6	23.0	19.1	16.4	12.8	9.0	6.7		
	10%	WB	3.0	3.1	4.7	8.1	13.3	15.0	17.5	16.3	14.3	10.9	7.4	4.9		
		MCDB	3.7	3.8	6.2	11.3	16.7	18.2	21.4	18.2	15.8	12.1	7.9	5.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.7	6.4	7.6	8.2	7.9	8.0	7.4	7.0	5.7	4.4	4.7		
		MCDBR	5.5	6.4	9.4	10.3	11.5	11.1	11.6	9.0	8.4	6.1	5.9	4.9		
	5% WB	MCWBR	5.2	4.5	5.9	5.7	5.7	5.3	4.9	4.7	5.4	4.9	5.6	5.0		
		MCWBR	4.9	4.9	7.9	9.7	9.5	8.8	9.4	6.9	6.6	5.3	5.5	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.314	0.334	0.374	0.367	0.354	0.376	0.372	0.343	0.330	0.295	0.253		
	taud		2.238	2.318	2.365	2.324	2.419	2.477	2.435	2.442	2.512	2.445	2.296	2.159		
	Ebn at Noon		534	703	805	829	863	881	849	825	796	682	508	419		
	Edh at Noon		43	67	86	106	103	99	101	94	75	60	41	32		
(44) All-Sky Solar Radiation	RadAvg		0.32	0.91	2.41	3.88	5.09	5.81	5.21	4.09	2.71	1.21	0.42	0.21		
	RadStd		0.06	0.17	0.24	0.54	0.52	0.39	0.61	0.28	0.31	0.21	0.10	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page